Newton Dividend Difference

$$f[x,x_0] = \frac{f(x) - f(x_0)}{x - x_0}$$

This can be written as

$$(x - x_0) f[x, x_0] = f(x) - f(x_0)$$

$$f(x) = f(x_0) + (x - x_0) f[x, x_0] \quad \text{— (1)}$$

Similarly

$$f[x, x_0, x_1] = \frac{f[x, x_0) - f[x_0, x_1]}{x - x_1}$$

We have

$$f[x, x_0) - f[x_0, x_1] = (x - x_1) f[x, x_0, x_1]$$

$$f[x, x_0) = f[x_0, x_1] + (x - x_1) f[x, x_0, x_1]$$

~~$f[x, x_0) = f$~~

Substitute $f[x, x_0)$ in equation

$$f(x) = f(x_0) + (x - x_0) f[x_0, x_1] + (x - x_1) f[x, x_0, x_1] (x - x_0) \quad \text{— (2)}$$

Now

$$f[x, x_0, x_1, x_2) = \frac{f[x, x_0, x_1) - f[x_0, x_1, x_2)}{x - x_2}$$

$$f[x, x_0, x_1) = f[x_0, x_1, x_2) + (x - x_2) f[x, x_0, x_1, x_2)$$

Put in eq (2)

$$f(x) = f(x_0) + (x - x_0) f[x_0, x_1] + (x - x_1)(x - x_0) f[x_0, x_1, x_2] + (x - x_1)(x - x_0)(x - x_2) f[x, x_0, x_1, x_2)$$